

#1a:

$P_{\text{rad}} = P_{\text{gas}} \leftarrow$ this defines the boundary
if $P_{\text{rad}} > P_{\text{gas}}$: then radiation dominates
if $P_{\text{gas}} > P_{\text{rad}}$: gas dominates

(2) $P_{\text{rad}} > P_{\text{gas}}$

plug in: $\frac{1}{3} a T^4 > \frac{R}{\mu} \rho T$

solve for T : $T^3 > \frac{3R}{a\mu} \rho$

$$T > \left(\frac{3R}{a\mu} \right)^{1/3} \rho^{1/3}$$

$$\left(\frac{3R}{a\mu} \right)^{1/3} = \left(\frac{3 (8.315 \times 10^7 \text{ erg/K})}{7.56 \times 10^{-15} \frac{\text{erg}}{\text{cm}^3 \text{K}^4}} \frac{1}{\mu} \right)^{1/3}$$
$$= \left(3.21 \times 10^7 \frac{\text{cm K}}{g^{1/3}} \right) \mu^{-1/3}$$

$$\mu_{\text{hydrogen}} \approx 1$$

$$T > 3.21 \times 10^7 \frac{\text{cm K}}{g^{1/3}} \rho^{1/3}$$

+5 for correct
to order of
magnitude

(ii) $T < 3.21 \times 10^7 \frac{\text{cm K}}{g^{1/3}} \rho^{1/3}$

+5 for correct to
order of magnitude

#16: (2)

$$M = 10 M_{\odot}$$

$$\left(\frac{M}{M_{\odot}}\right)^{0.7} = \left(\frac{R}{R_{\odot}}\right)$$

$$R = 5.01 R_{\odot}$$

$$\rho = \frac{M}{\frac{4}{3}\pi R^3} = 0.02 \frac{M_{\odot}}{R_{\odot}^3}$$

$$\rho_{\text{cgs}} = 0.112 \frac{\text{g}}{\text{cm}^3}$$

+1 for trying
+1 for correct

$$M = 150 M_{\odot}$$

$$\left(\frac{M}{M_{\odot}}\right)^{0.7} = \frac{R}{R_{\odot}}$$

$$R = 33.36 R_{\odot}$$

$$\rho = 0.00096 \frac{M_{\odot}}{R_{\odot}^3}$$

$$\rho_{\text{cgs}} = 0.0057 \frac{\text{g}}{\text{cm}^3}$$

+1 for trying
+1 for correct

$$(ii) T_{\text{ave}} = \frac{\alpha}{3} \frac{\mu m_u}{k} \frac{GM}{R}$$

$$\alpha = 1, \mu = 0.5, m_u = 1.66 \times 10^{-27} \text{ kg}$$

$$T_{\text{ave}, 10 M_{\odot}} = 2.12 \times 10^{-5} \frac{\text{kg K}}{\text{J}} \frac{G (10 M_{\odot})}{5 R_{\odot}}$$

$$\boxed{\begin{aligned} T_{\text{ave}, 10 M_{\odot}} &= 8.1 \times 10^6 \text{ K} \\ T_{\text{ave}, 150 M_{\odot}} &= 1.82 \times 10^7 \text{ K} \end{aligned}}$$

+1 each for trying
+1 each for correct

$$(iii) \frac{T_{\text{ave}, 10 M_{\odot}}}{\rho_{10 M_{\odot}}^{1/3}} = \frac{8.1 \times 10^6 \text{ K}}{(0.112 \text{ g/cm}^3)^{1/3}} = 1.68 \times 10^7 \frac{\text{cm K}}{\text{g}^{1/3}}$$

$$\rightarrow T_{\text{ave}, 10 M_{\odot}} < 3.21 \times 10^9 \frac{\text{cm K}}{g^{1/3}} \rho_{10 M_{\odot}}$$

$\Rightarrow$ gas dominates for $10 M_{\odot}$

$$\frac{T_{\text{ave}, 150 M_{\odot}}}{(\rho_{150 M_{\odot}})^{1/3}} = 1.02 \times 10^8 \frac{\text{cm K}}{g^{1/3}}$$

$\Rightarrow$ radiation dominates for $150 M_{\odot}$

+1 each for trying

+1 each for correct

#2 : +10 for trying

+10 for coherent